Part I: Practice Workbook 11/02 — KEY

ML & FinTech 115-1 · Sessions 1–2 · work alone, pen and calculator. We go over each page in class.

Page A. Confusion matrix (ISLP §4.4.2)

Question 1 (6 min). ISLP Table 4.4: a classifier on the Default data, 10,000 people, threshold 0.5.

	true: no default	**true: default**	**total**
predicted: no default	9644	252	9896
predicted: default	23	81	104
total	9667	333	10000

Write TP, FP, FN, TN. Compute accuracy, sensitivity (TPR), specificity (TNR), precision.

Procedure. Positive = default. TP: predicted default, true default = 81. FP: predicted default, true no default = 23. FN: predicted no default, true default = 252. TN: predicted no default, true no default = 9644.
Accuracy = (TP + TN) / total = (81 + 9644) / 10000 = 0.973.
Sensitivity = TP / (TP + FN) = 81 / (81 + 252) = 0.243.
Specificity = TN / (TN + FP) = 9644 / (9644 + 23) = 0.998.
Precision = TP / (TP + FP) = 81 / (81 + 23) = 0.779.

Question 2 (3 min). A rule says "no one defaults". Compute its accuracy, sensitivity and specificity. What does accuracy hide here?

Procedure. The rule predicts "no default" for everyone, so TP = 0, FP = 0, FN = 333, TN = 9667.

Accuracy = (0 + 9667) / 10000 = 0.9667.

Sensitivity = 0 / 333 = 0.

Specificity = 9667 / 9667 = 1.

Gap to the model: 0.9725 − 0.9667 = 0.0058.
Conclusion: the model beats it by only 0.0058. Accuracy hides that the model misses 252 of 333 defaulters: with few defaulters, accuracy mostly counts the non-defaulters.

Question 3 (6 min). ISLP Table 4.5: the same classifier at threshold 0.2 gives TP 195, FN 138, FP 235, TN 9432. Compute sensitivity and specificity. If a missed default costs 10 times a refused good customer, compare the cost FP + 10 × FN at thresholds 0.5 and 0.2. Which threshold do you choose?

Procedure.

Sensitivity = 195 / (195 + 138) = 0.586.

Specificity = 9432 / (9432 + 235) = 0.976.

Accuracy = (195 + 9432) / 10000 = 0.963.
Cost at 0.5: 23 + 10 × 252 = 23 + 2520 = 2543.
Cost at 0.2: 235 + 10 × 138 = 235 + 1380 = 1615.
Conclusion: choose 0.2: accuracy falls a little, cost falls a lot.

Page B. ROC curve and AUC

Eight applicants, sorted by predicted probability: 4 defaulted (y = 1), 4 did not (y = 0).

$\hat{p}$	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2
y	1	1	0	1	0	1	0	0

Question 1 (5 min). At threshold 0.5 and at 0.8 (predict 1 when $\hat{p} \geq$ threshold): write TP, FP, FN, TN, then TPR = TP/P and FPR = FP/N.

Procedure. Count P = 4 (y = 1) and N = 4 (y = 0).

Threshold 0.5: the first 5 applicants are predicted 1 ($\hat{p} \geq 0.5$).

Their y are 1, 1, 0, 1, 0, so TP = 3, FP = 2. The last 3 are predicted 0, with y = 1, 0, 0, so FN = 1, TN = 2.

TPR = 3/4 = 0.75; FPR = 2/4 = 0.50.

Threshold 0.8: the first 2 are predicted 1.

Their y are 1, 1, so TP = 2, FP = 0. Of the other 6, two have y = 1, so FN = 2, TN = 4.

TPR = 2/4 = 0.50; FPR = 0/4 = 0.

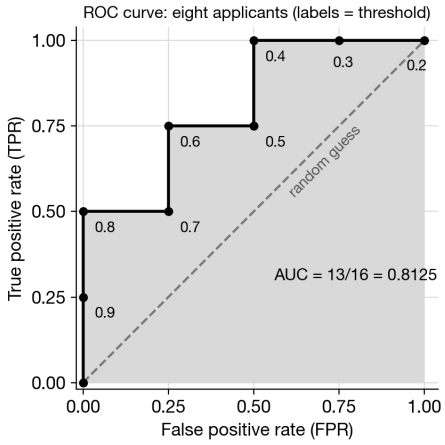
Question 2 (5 min, on the board together). Lower the threshold one applicant at a time and plot (FPR, TPR), starting from (0, 0). Draw the diagonal too. What does the diagonal mean?

Procedure. Start at (0, 0). Go down the list: y = 1 moves TPR up by 1/4 = 0.25; y = 0 moves FPR right by 1/4 = 0.25.

Points: (0,0) → (0.00,0.25) → (0.00,0.50) → (0.25,0.50) → (0.25,0.75) → (0.50,0.75) → (0.50,1.00) → (0.75,1.00) → (1.00,1.00).

Conclusion: the diagonal is a classifier guessing at random.

Threshold	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2
FPR	0	0	0.25	0.25	0.50	0.50	0.75	1
TPR	0.25	0.50	0.50	0.75	0.75	1	1	1



Question 3 (5 min). AUC is the fraction of (defaulter, non-defaulter) pairs in which the defaulter has the higher $\hat{p}$. Count the pairs. Does AUC depend on the threshold?

Procedure. Total pairs = 4 × 4 = 16. For each defaulter, count the non-defaulters below it: 4 + 4 + 3 + 2 = 13. AUC = 13/16 = 0.8125.

Conclusion: no, it summarises all thresholds, so it cannot choose one. Choosing needs costs.

Page C. Logistic regression with more than two classes (ISLP §4.3.5)

Three classes for a client: 1 = pays on time, 2 = late, 3 = default. For one client the softmax scores are $\eta_1 = 1$, $\eta_2 = 0$, $\eta_3 = -1$. $e \approx 2.718$, $e^2 \approx 7.39$.

Question 1 (5 min). Softmax coding: $P(Y = k \mid x) = e^{\eta_k} / (e^{\eta_1} + e^{\eta_2} + e^{\eta_3})$. Compute the three probabilities. Check that they sum to 1.

Procedure.

$e^1 = 2.718$, $e^0 = 1$, $e^{-1} = 1/2.718 = 0.368$.

Denominator = 2.718 + 1 + 0.368 = 4.086.

P(Y=1) = 2.718/4.086 = 0.665. P(Y=2) = 1/4.086 = 0.245. P(Y=3) = 0.368/4.086 = 0.090.

Check: 0.665 + 0.245 + 0.090 = 1.

Question 2 (5 min, at home). Baseline coding with class 3 as baseline: subtract η_3 from every score. Write the new scores, compute the probabilities again. What changed and what did not? What is log[P(Y=1|x)/P(Y=3|x)]?

Procedure. Subtract $\eta_3 = -1$: new scores 1 + 1 = 2, 0 + 1 = 1, −1 + 1 = 0.

Denominator = $e^2 + e^1 + e^0 = 7.39 + 2.718 + 1 = 11.107$.

P(Y=1) = 7.39/11.107 = 0.665. P(Y=2) = 2.718/11.107 = 0.245. P(Y=3) = 1/11.107 = 0.090.

Why the same: subtracting a constant multiplies top and bottom by the same $e^{(-\eta_3)}$.
Log-odds:

log[P(Y=1|x)/P(Y=3|x)] = log(0.665/0.090) = 2,
the baseline score of class 1.
Conclusion: scores change, probabilities do not.

Question 3 (5 min, at home). The softmax slopes on x (months late, say) are 0.5, 0.2, -0.3 for classes 1, 2, 3. Write the slopes in baseline coding. One more unit of x multiplies the odds of class 1 against class 3 by how much? Can you read the softmax slope 0.5 alone as an effect on class 1?

Procedure.
Subtract the class 3 slope: 0.5 − (-0.3) = 0.8; 0.2 − (-0.3) = 0.5; -0.3 − (-0.3) = 0.

The log-odds of class 1 against class 3 rises by 0.8 per unit of x, so the odds are multiplied by $e^{0.8} = 2.23$.

Conclusion: no, a multinomial coefficient means something only against a baseline.